

Specifications



Material		
Item	Specification	Fill Capacity
Motorcraft® Gasket Maker TA-16	WSK- M2G348-A5	—
Motorcraft® High Performance Engine RTV Silicone TA-357	WSE- M4G323-A6	—
Motorcraft® Metal Surface Prep Wipes ZC-31-B	—	—
Motorcraft® SAE 5W-30 Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-30 Super Premium Motor Oil (Canada) XO-5W30-QSP (US); CXO-5W30- LSP12 (Canada)	WSS- M2C946-A	5.6L (6.0 qts.) includes oil filter change
Motorcraft® Orange Concentrated Antifreeze/Coolant VC-3-B (US); CVC-3-B2 (Canada)	WSS- M97B44-D	—
Motorcraft® Silicone Gasket Remover ZC-30-A	—	—
Motorcraft® Thread Sealant with PTFE TA-24-B	WSK- M2G350-A2	—
Motorcraft® Threadlock 262 TA-26	WSK- M2G351-A6	—

General Specifications

Item	Specification
Engine	
Displacement	3.5L Gasoline Turbocharged Direct Injection (GTDI) (214 CID)
No. cylinders	6
Bore/stroke	92.5/86.7 mm (3.641/3.413 in)
Fire order	1-4-2-5-3-6
Spark plug	12405
Spark plug gap	0.70-0.95 mm (0.028-0.037 in)
Oil pressure	Minimum 30 psi @ 1,500 rpm with engine at normal operating

Item	Specification
	temperature
Compression ratio	10.0:1
Engine weight (without accessory drive components)	189.01 kg (416.7 lb)
Cylinder Head and Valve Train	
Cylinder head gasket surface flatness	Flat within 0.08 mm (0.003 in) length end to end; area 150 mm (5.9 in) x 150 mm (5.9 in) (or full width) should be less than 0.05 mm (0.002 in)
Combustion chamber volume	55.84 cc (3.41 CI)
Valve tappet clearance — intake	0.15-0.25 mm (0.006-0.01 in)
Valve tappet clearance — exhaust	0.360-0.460 mm (0.0142-0.0181 in) Engine must be at room temperature before measuring
Valve guide bore inner diameter	5.519-5.549 mm (0.217-0.218 in)
Valve stem diameter — intake	5.479-5.497 mm (0.2157-0.2164 in)
Valve stem diameter — exhaust	5.466-5.484 mm (0.2151-0.2159 in)
Valve stem-to-guide clearance — intake	0.022-0.070 mm (0.0008-0.0027 in)
Valve stem-to-guide clearance — exhaust	0.035-0.083 mm (0.0014-0.0033 in)
Valve head diameter — intake	36.82-37.18 mm (1.44-1.46 in)
Valve head diameter — exhaust	30.82-31.18 mm (1.21-1.22 in)
Valve face runout	0.05 mm (0.0001 in)
Valve face angle	45.75-45.25 degrees
Valve seat width — intake	1.3-1.6 mm (0.051-0.063 in)
Valve seat width — exhaust	1.7-2.0 mm (0.067-0.079 in)
Valve seat runout	0.04 mm (0.0001 in) MAX
Valve seat angle	44.5-45.5 degrees
Valve spring free length (approx.)	55.16 mm (2.17 in)
Valve spring compression pressure (N @ spec. length)	465 N @ 28.0 mm (105 lb @ 1.10 in)
Valve spring installed height	37.0 mm (1.45 in)
Valve spring installed height pressure (N @ spec. length)	265 N @ 37.0 mm (60 lb @ 1.45 in)
Valve spring installed pressure — service limit	10% force loss @ specified height
Camshaft	
Theoretical valve lift @ 0 lash	9.49 mm (0.373 in)
Lobe lift — intake	9.49 mm (0.373 in)
Lobe lift — exhaust	9.49 mm (0.373 in)
Allowable lobe lift loss	0.062 mm (0.0024 in)
Camshaft journal bore inside diameter — 1st journal	39.0375-39.0625 mm (1.537-1.538 in)
Camshaft journal bore inside diameter — intermediate journals	25.9875-26.0125 mm (1.023-1.024 in)
Camshaft bearing outside diameter — 1st journal	38.990-39.010 mm (1.535-1.536 in)

Item	Specification
Camshaft bearing outside diameter — intermediate journals	25.937-25.963 mm (1.021-1.022 in)
Camshaft journal-to-bearing clearance, 1st journal — service limit	0.0725 mm (0.0029 in) MAX
Camshaft journal-to-bearing clearance, intermediate journals — service limit	0.0755 mm (0.0029 in) MAX
Runout	0.040 mm (0.0015 in) MAX
End play — standard	0.032-0.170 mm (0.0012-0.0066 in)
End play — service limit	0.190 mm (0.00748 in) MAX
Cylinder Block	
block main bore roundness	0.008 mm (0.003 in)
Cylinder bore diameter — grade 1	92.500-92.520 mm (3.641-3.642 in)
Cylinder bore roundness	0.013 mm (0.0005 in)
Cylinder bore taper	0.010 mm (0.0004 in) per 25.4 mm (1.000 in)
Main bearing bore inside diameter	72.400-72.424 mm (2.8503-2.8513 in)
Head gasket surface flatness	Flat within 0.150 mm (0.005 in) overall 0.050 mm (0.001 in) per 150 mm (5.905 in) x 150 mm (5.905 in); 0.025 mm (0.0009 in) per 25 mm (0.984 in) x 25 mm (0.984 in)
Crankshaft	
Main bearing journal diameter	67.5 mm (2.657 in)
Main bearing journal maximum taper	0.004 mm (0.00015 in)
Main bearing journal maximum out-of-round	0.006 mm (0.00023 in)
Main bearing journal-to-cylinder block clearance	0.026-0.041 mm (0.0010-0.0016 in)
Connecting rod journal diameter	55.983-56.003 mm (2.204-2.205 in)
Connecting rod journal maximum taper	0.004 mm (0.00015 in)
Connecting rod journal maximum out-of-round	0.006 mm (0.00023 in)
Crankshaft maximum end play	0.101-0.291 mm (0.0039-0.0114 in)
Piston and Connecting Rod	
Piston diameter — single grade	92.476-92.490 mm (3.6407-3.6413 in)
Piston-to-cylinder bore clearance	0.010 to 0.044 mm (0.0003-0.0017 in)
Piston ring end gap — compression (top, gauge diameter)	0.17-0.27 mm (0.0067-0.0106 in)
Piston ring end gap — compression (bottom, gauge diameter)	0.30-0.55 mm (0.0118-0.0216 in)
Piston ring end gap — oil ring (steel rail, gauge diameter)	0.15-0.45 mm (0.0059-0.0177 in)
Piston ring groove width — compression (top)	1.230-1.25 mm (0.0484-0.0492 in)
Piston ring groove width — compression (bottom)	1.530-1.55 mm (0.0602-0.0610 in)
Piston ring groove width — oil ring	2.53-2.55 mm (0.0996-0.1003 in)
Piston ring width — upper compression ring	1.17-1.19 mm (0.0460-0.0468 in)

Item	Specification
Piston ring width — lower compression ring	1.47-1.49 mm (0.0578-0.0586 in)
Piston ring-to-groove clearance (upper and lower compression rings)	0.040-0.080 mm (0.0014-0.0031 in)
Piston pin bore diameter	23.004-23.008 mm (0.9056-0.9058 in)
Piston pin diameter	22.997-23.000 mm (0.9053-0.9055 in)
Piston pin length	2.343 in (59.512 mm)
Piston pin-to-piston fit	0.004-0.011 mm (0.00015-0.00043 in)
Piston-to-connecting rod clearance	3.3 mm (0.13 in)
Connecting rod-to-pin clearance — standard	0.007-0.022 mm (0.0002-0.0009 in)
Connecting rod pin bore diameter	23.007-23.019 mm (0.905-0.906 in)
Connecting rod length (center-to-center)	152.68 mm (6.01 in)
Connecting rod maximum allowed bend	0.038 mm (0.0014 in)
Connecting rod maximum allowed twist	0.050 mm (0.0019 in)
Connecting rod bearing bore diameter — grade 1	59.866-59.872 mm (2.3569-2.3571 in)
Connecting rod bearing bore diameter — grade 2	59.873-59.879 mm (2.3572-2.3574 in)
Connecting rod bearing bore diameter — grade 3	59.880-59.886 mm (2.3574-2.3577 in)
Connecting rod bearing-to-crankshaft clearance	0.020-0.054 mm (0.0007-0.0021 in)
Connecting rod side clearance (assembled to crank) — standard	0.175-0.425 mm (0.0068-0.0167 in)
Connecting rod side clearance (assembled to crank) — service limit	0.175-0.425 mm (0.0068-0.0167 in)

Torque Specifications

Description	Nm	lb-ft	lb-in
A/C compressor bolt	25	18	—
A/C compressor stud	9	—	80
A/C tube nut	15	—	133
Accessory drive belt tensioner bolts	25	18	—
Axle shaft housing carrier bushing bolt	115	85	—
Bellhousing-to-engine bolts and stud bolts	48	35	—
Bellhousing-to-oil pan bolts	48	35	—
Block coolant drain plug — RH ^a	—	—	—
Block coolant drain plug — LH ^b	—	—	—
Block oil gallery plug — LH ^c	48	35	—
Block oil gallery plug — rear ^c	85	63	—
Block oil gallery plug — RH ^c	32	24	—

Description	Nm	lb-ft	lb-in
Charge Air Cooler (CAC) tube bracket nuts	6	—	53
CAC tube clamps	5	—	44
CAC -to-Throttle Body (TB) pipe clamp	5	—	44
Camshaft bearing cap assembly bolts ^d	—	—	—
Camshaft bearing cap bolts ^d	—	—	—
Camshaft Position (CMP) sensor bolts (LH cylinder head)	10	—	89
CMP sensor bolts (RH valve cover)	7	—	62
CMP sensor heat shield bolt	10	—	89
CMP sensor heat shield nut	10	—	89
Catalytic converter-to-exhaust manifold nuts	45	35	—
Catalytic converter-to-intermediate pipe bolts ^d	—	—	—
Connecting rod cap bolts ^d	—	—	—
Coolant pump bolts ^d	—	—	—
Coolant pump pulley bolts	24	18	—
Crankshaft rear seal retainer plate bolts ^d	—	—	—
Crankshaft pulley bolt ^d	—	—	—
Crankshaft Position (CKP) sensor bolt	10	—	89
CKP sensor heat shield bolt	10	—	89
CKP sensor heat shield nut	10	—	89
Crossmember brace nuts and bolts	90	66	—
Cylinder head bolts ^d	—	—	—
Cylinder head M6 bolt	10	—	89
Cylinder Head Temperature (CHT) sensor	11	—	97
Engine appearance cover bracket bolts	10	—	89
Engine coolant crossover assembly bolt ^d	—	—	—
Engine front cover bolts ^d	—	—	—
Engine support insulator bolts	175	129	—
Engine support insulator bracket bolts - LH side	63	46	—
Engine support insulator bracket bolts - RH side, standard payload package	63	46	—
Engine support insulator bracket bolts - RH side, heavy duty payload package	77	53	—
Engine support insulator nuts	175	129	—
Engine support insulator studs	30	22	—
Engine support insulator through bolts	350	258	—
Engine oil filter ^e	—	—	—
Engine-to-bellhousing bolt	45	35	—
Engine Oil Pressure (EOP) switch ^d	—	—	—
Evaporative Emission (EVAP) canister purge valve bolt	10	—	89

Description	Nm	lb-ft	lb-in
Exhaust manifold nuts ^d	—	—	—
Exhaust manifold studs	12	—	106
Flexplate bolts ^d	—	—	—
Fuel injection pump mounting plate bolt ^d	—	—	—
Fuel rail bolts ^d	—	—	—
Generator B+ terminal nut	17	—	150
Generator nut and bolt	48	35	—
Generator stud	8	—	71
Ground strap bolt	10	—	89
Ground stud bolt ^d	—	—	—
Hood bolts	12	—	106
High pressure fuel tube bracket bolt ^d	—	—	—
High pressure fuel tube bracket nut	8	—	71
High pressure fuel tube flare nut ^d	—	—	—
Ignition coil-on-plug bolts ^d	—	—	—
Intake manifold bolts ^d	—	—	—
Knock Sensor (KS) bolts	20	—	177
Lower front axle carrier mounting bushing bolt	115	85	—
Main bearing bolts ^d	—	—	—
Main bearing side bolts ^d	—	—	—
Main bearing cap support brace bolts ^d	—	—	—
Oil cooler insert bolt	58	43	—
Oil filter adapter bolts ^d	—	—	—
Oil pan bolts ^d	—	—	—
Oil pan drain plug	27	20	—
Oil pump bolts	10	—	89
Oil pump screen and pickup tube bolts	10	—	89
Piston oil cool valves	4	—	35
Primary timing chain tensioner bolts	10	—	89
Skid plate bolts	48	35	—
Radio ignition interference capacitor bolt	10	—	89
Spark plugs	15	—	133
Thermostat housing bolts ^d	—	—	—
Timing chain gear bolts ^d	—	—	—
Timing chain guide bolts	10	—	89
Timing chain guide pin ^f	—	—	—

Description	Nm	lb-ft	lb-in
Timing chain tensioner arm guide pin ^f	—	—	—
Torque converter nuts	44	32	—
Transmission cooler tube bracket bolt	12	—	106
Transmission cooler tube bracket nut	12	—	106
Transmission crossmember nuts	103	75	—
Turbocharger intake tube clamp	5	—	44
Turbocharger bracket-to-engine bolts	10	—	89
Turbocharger oil return tube assembly bolts ^g	—	—	—
Turbocharger-to-mounting bracket bolt	28	—	21
Turbocharger-to-exhaust manifold bolts	32	24	—
Vacuum pump bolts ^d	—	—	—
Wiring harness retainer stud bolt	10	—	89
Valve cover bolts and stud bolts ^d	—	—	—
Valve train oil tube bolts ^d	—	—	—
Variable Camshaft Timing (VCT) bolts ^d	—	—	—
VCT oil control solenoid bolts ^d	—	—	—
Wiring harness retainer stud bolt	10	—	89

^aTighten to 20 Nm (177 lb-in) plus an additional 180 degrees.

^bTighten to 10 Nm (89 lb-in) plus an additional 720 degrees

^cDiscard original plug and O-ring seal. Install a new plug and O-ring seal to specified torque, no sealer or lubricant required.

^dRefer to the procedure in this section.

^eTighten to 5 Nm (44 lb-in) plus an additional 180 degrees.

^f20 Nm (177 lb-in) plus an additional 60 degrees

^gRefer to the assembly procedure in this section.